1.4 Exercises

For each of the following problems, identify the problem inputs, processes, and problem outputs. Then write pseudocode and create flowchart for each solution.

a) A program will change a temperature reading from Celsius to Fahrenheit. Prompt the user to enter the temperature in degree Celsius and display the temperature in Fahrenheit. (Fahrenheit = 32 + (9*Celsius/5))

   Analysis :

   Input : Celsius Temp, CELCIUS

   Process:

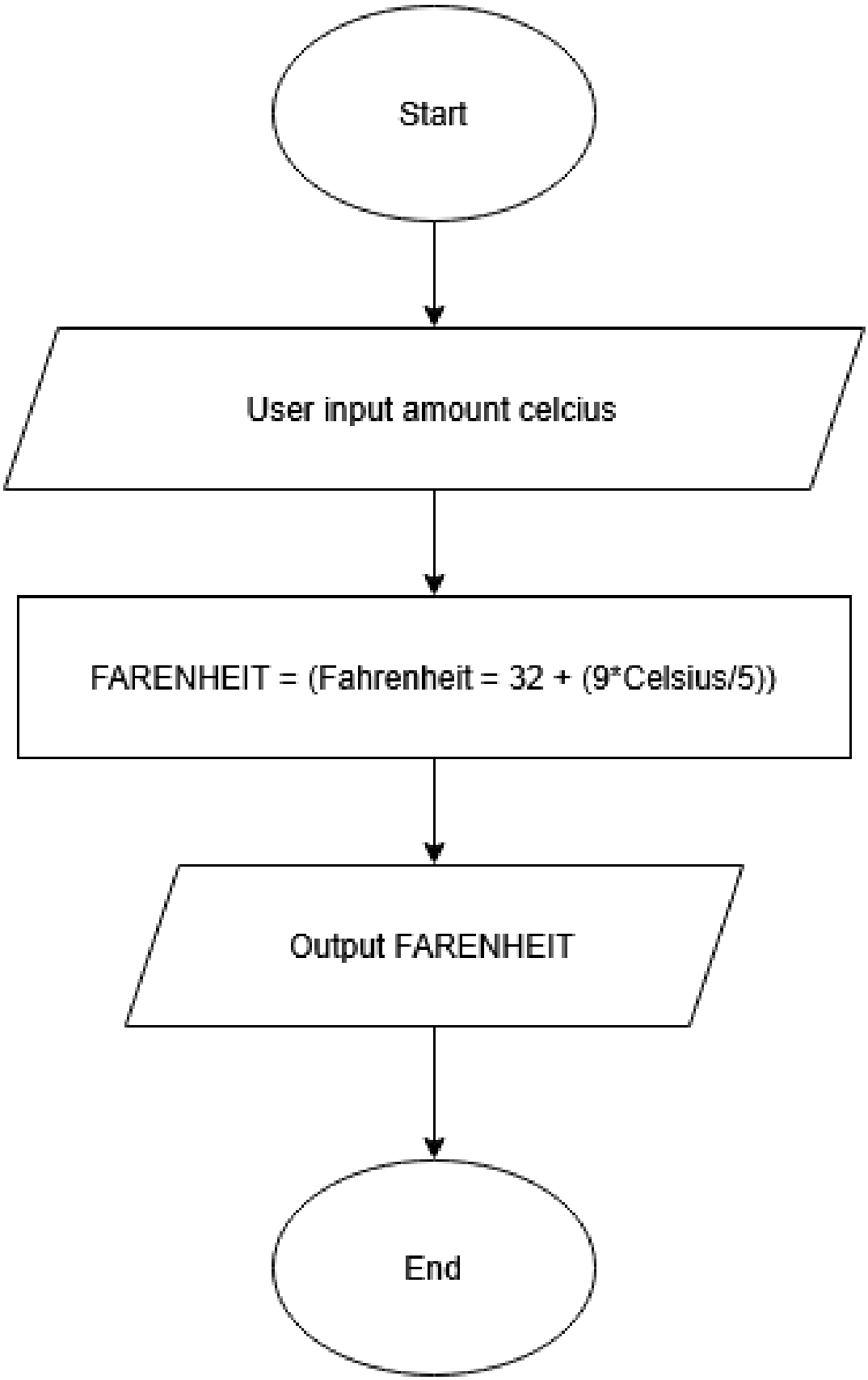
   i. Accept Celcius
   ii. FAHRENHEIT = 32 + (9*CELCIUS/5)
   iii. Display FAHRENHEIT

   Output : FARENHEIT

   Psedocode :

1. Start
2. Input CELCIUS
3. FAHRENHEIT = 32 + (9*CELCIUS/5)
4. Display FAHRENHEIT
5. End

FlowChart:


Start
User input amount celcius
FARENHEIT = (Fahrenheit = 32 + (9*Celsius/5))
Output FARENHEIT
End

b) Let a runner insert how many steps he has run in one minute, and the running duration in minutes and seconds. Find how many miles he has run. Assume each of his step length is 2.5 feet. (1 mile = 5280 feet)

Analysis:
Input:

i. Steps per minutes,

Prosess:

- Calculate the Distance in miles
  i. miles = (steps x 2.5)/5820 miles

Output:

i. Distance in miles

Psedocode:

Start

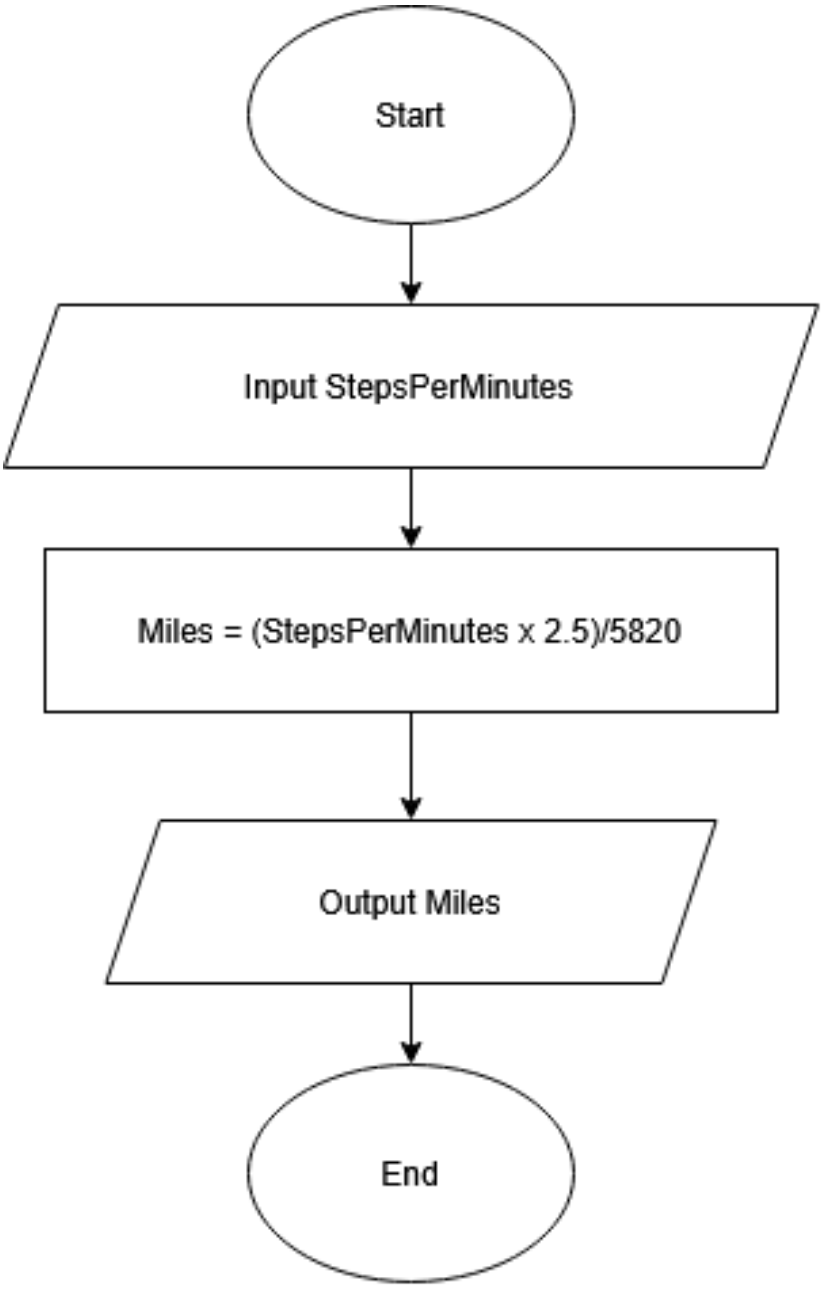
Input StepsPerMinutes

Miles = (StepsPerMinutes x 2.5)/5820 miles

Output Miles

End

FlowChart:


Start
Input StepsPerMinutes
Miles = (StepsPerMinutes x 2.5)/5820
Output Miles
End

c) Accept two numbers and find their sum and difference. If the sum is more than the difference, display the sum otherwise display the difference.

Analysis :
input :

i. FirstNum
ii. SecNum

Process:
-

i. Sum = FirstNum + SecNum
ii. Diff = FirstNum – SecNum
iii. Check if Sum is more than Diff
iv. Display Result

Output: Message " Yes, Sum Is More Than Difference" or "No, Difference Is More Than Sum"

Pseudocode:

```
Start

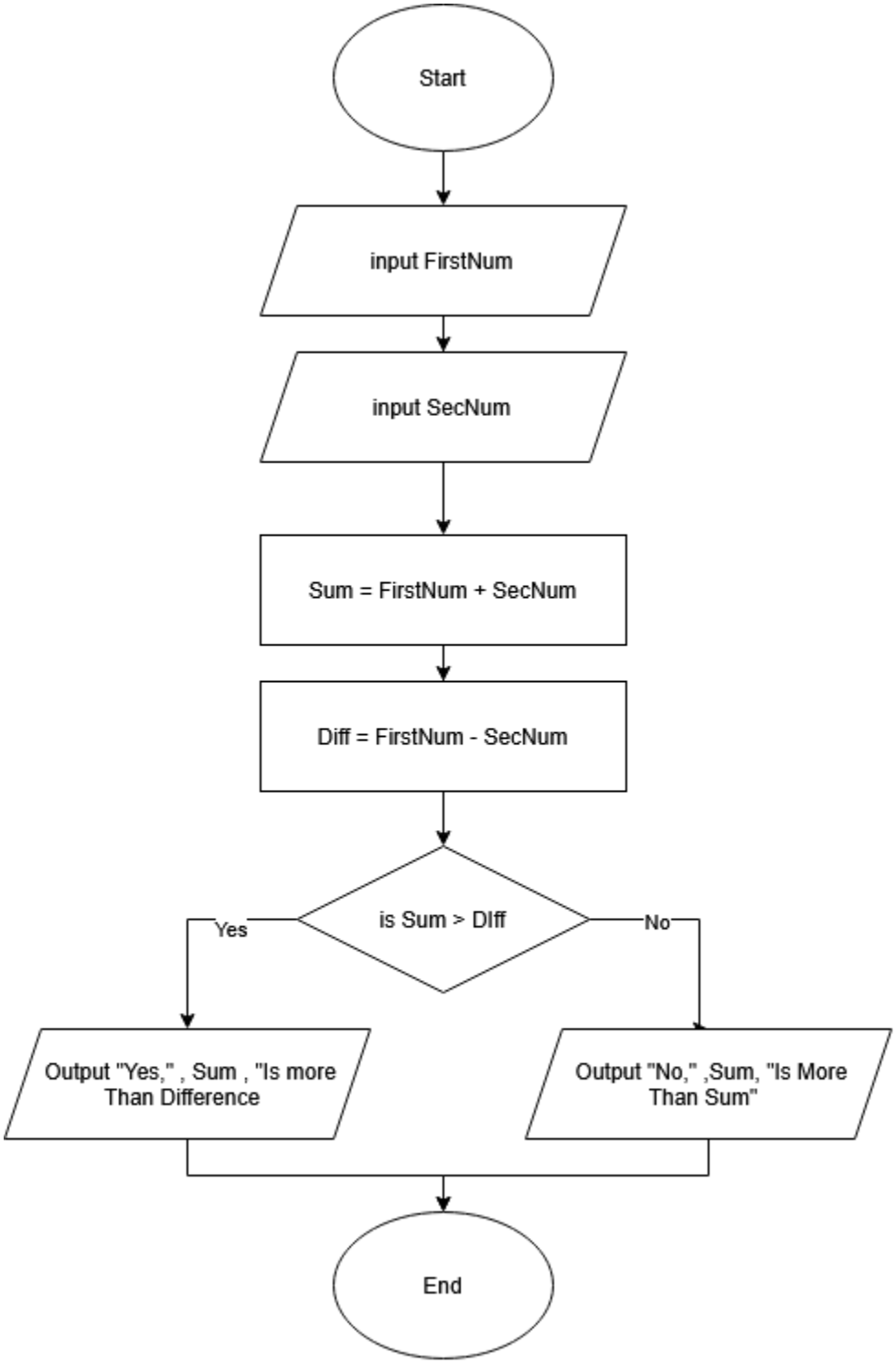
Input FirstNum
Input SecNum

    Sum = FirstNum + SecNum
    Diff = FirstNum – SecNum

    IF sum> Difference
        Output "Yes,", sum, "Is More Than Difference"
    Else
        Output "No,", sum, "Is More Than Sum"
    endIF

End
```

Flowchart :


Start
input FirstNum
input SecNum
Sum = FirstNum + SecNum
Diff = FirstNum - SecNum
is Sum > DIff
Yes
No
Output "Yes," , Sum , "Is more Than Difference
Output "No," ,Sum, "Is More Than Sum"
End

d) This is a program to calculate and display the body mass index (BMI). The BMI is to determine whether a person is overweight, underweight or has an optimal weight for his/her height. A person's BMI is calculated using the formula BMI = weight / height2 . A sedentary person's weight is considered to be optimal if his/her BMI is between 18.5 and 25. If the BMI is less than 18.5, the person is considered to be underweight. If the BMI value is greater than 25, the person is considered to be overweight.To determine the BMI, let the organiser enters the total number of people participating in this program. Then let each participant enters his/her weight (in kg) and height (in meters); and as a result, the participant gets a message indicating whether he/she has optimal weight, is underweight, or is overweight. Repeat the process for all participants.

Analysis :
Ch
Input :

i. totalParticipants
ii. weight
iii. height

Process :

i. ReadTotalParticipants
ii. Loop from 1 to ReadTotalParticipants
    - Read Weight and Height
    - Calculate BMI :BMI weight/Height^2
iii. Display BMI By Category
    - if BMI, <18.5 is Underweight
    - if BMI, < 25 is Optimal Weight
    - if BMI, >26 is Overweight

Output:

i. Message BMI With By Category

Psedocode :

```
Start

Input totalParticipants

For counter = 1 to totalParticipants
        input weight
        input height
BMI = weight / (height* Height)

        If BMI < 18.5

                Output “BMI: “, BMI, “ – Status: Underweight”

        Else If Bmi <= 25

                 Output “BMI: “, BMI, “ – Status: Optimal weight”

        Else

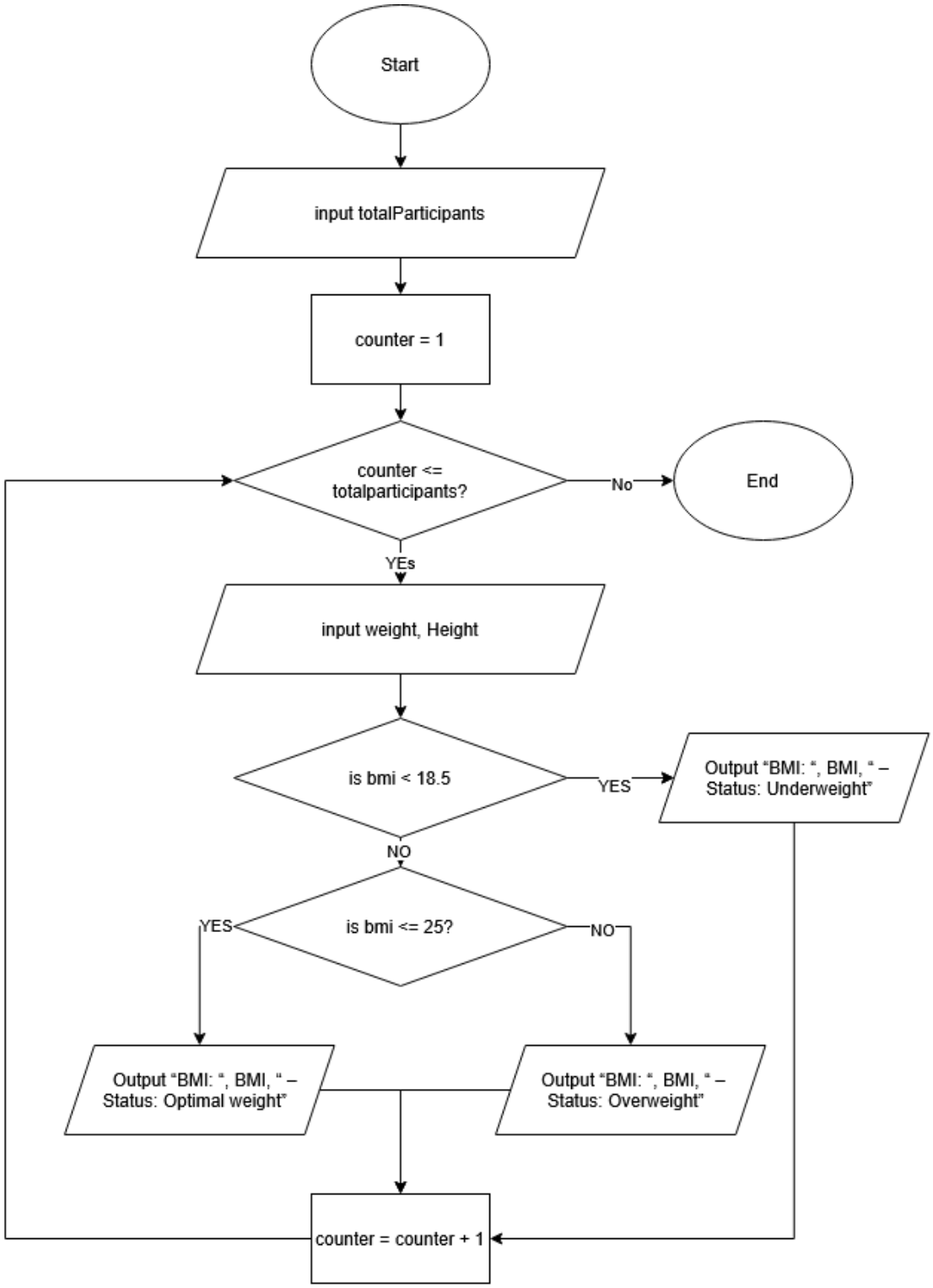
                Output “BMI: “, BMI, “ – Status: Overweight”

        EndIf

End For

End
```

FlowChart:


Start
input totalParticipants
counter = 1
counter <= totalparticipants?
No
End
YEs
input weight, Height
is bmi < 18.5
YES
Output "BMI: ", BMI, " – Status: Underweight"
NO
is bmi <= 25?
YES
NO
Output "BMI: ", BMI, " – Status: Optimal weight"
Output "BMI: ", BMI, " – Status: Overweight"
counter = counter + 1

e) Insert a few integer numbers between -100 and 100 (insert the value 0 to stop). Determine how many negative numbers and how many positive numbers have been inserted, the smallest number and the biggest number, their total and average. Print out what have been determined. (Consider to use the top-down approach in your design).

Analysis:

Input:

i. Number

Process :

1. Set the Variables:
   i. posCount =0,
   ii. negCount = 0
   iii. total = 0
   iv. count = 0
   v. smallest = 100
   vi. biggest = 100
2. Read First Integer “Number”
3. While number != 0:
   i. Range: -100 ≤ number ≤ 100
   ii. Increment: count = count +1
   iii. Update total: total = total + number
   iv. Check sign:
      - If number > 0, increment posCount = posCount +1
      - If number < 0, increment negCount = negCount +1
   v. Update:
      - number < smallest, Update smallest = number
      - number > biggest, update biggest = number
   vi. Read next “number”
4. If count > 0 :
   i. average = total/count

Output:

Output posCount
Output negCount
Output Smallest
Output Biggest
Output Total
Output Average

Pseudocode :

```
Start

posCount =0,
negCount = 0
total = 0
count = 0
smallest = 100
biggest = 100

input Number

While number != 0

        If number >= -100 AND number <= 100
                Count = count + 1
                total = total + number

        If number > 0
                posCount = posCount +1

        Else
                negCount = negCount +1
        EndIf

        If number > biggest
                biggest = number
        endif

else
        Output" Invalid input! Number Must be Between -100 and 100."
endif
```

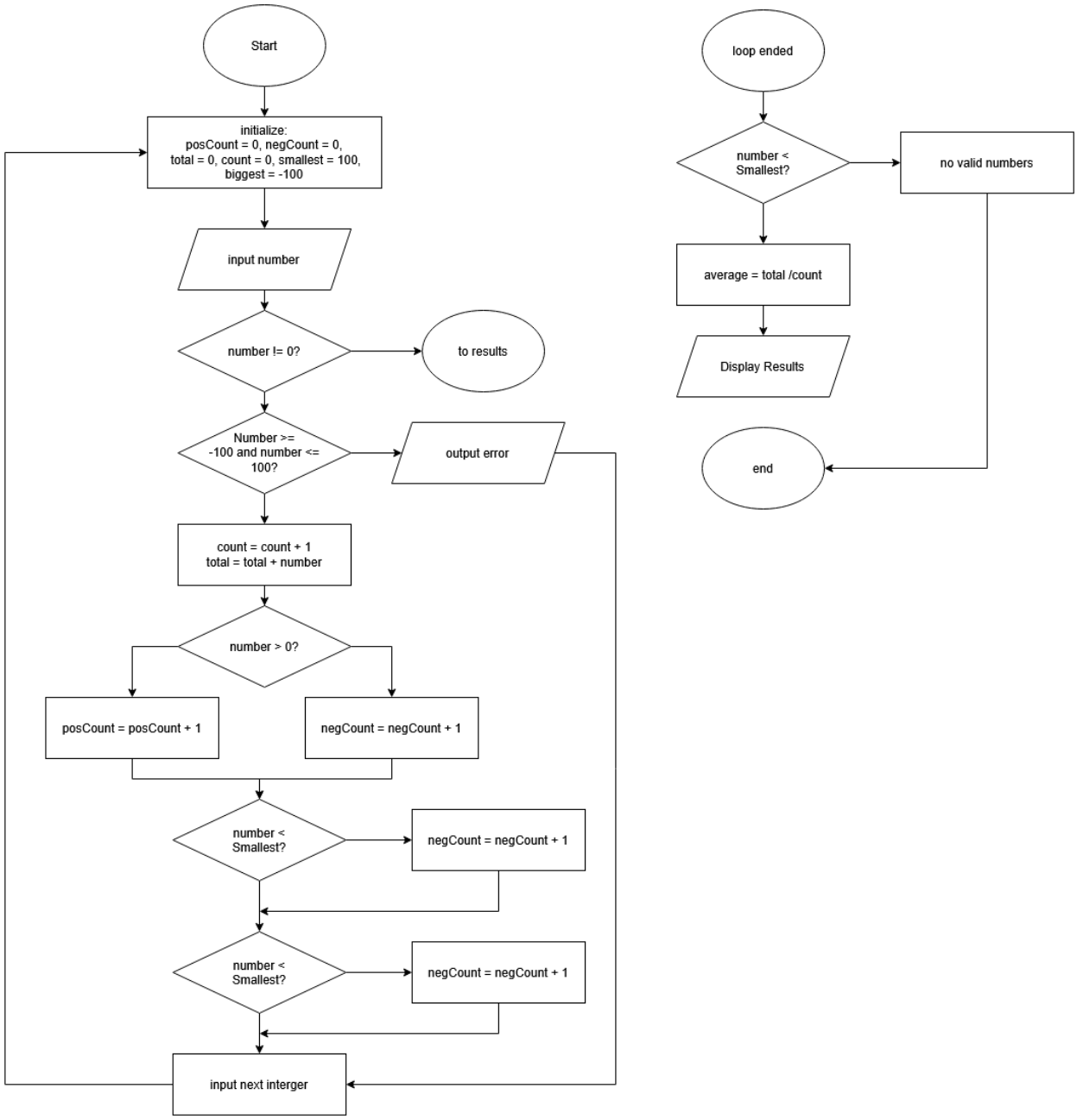
```
        input next number
Endwhile


If count > 0
        average = total / count

        Output "Positive numbers count: ", posCount
        Output "Negative numbers count: ", negCount
        Output "Smallest number: ", smallest
        Output "Biggest number: ", biggest
        Output "Total sum: ", total
        Output "Average: ", average
Else

        Output “No valid numbers were entered”
endIf
End
```

flowchart :


Start
initialize:
posCount = 0, negCount = 0,
total = 0, count = 0, smallest = 100,
biggest = -100
input number
number != 0?
to results
Number >=
-100 and number <=
100?
output error
count = count + 1
total = total + number
number > 0?
posCount = posCount + 1
negCount = negCount + 1
number <
Smallest?
negCount = negCount + 1
number <
Smallest?
negCount = negCount + 1
input next interger
loop ended
number <
Smallest?
no valid numbers
average = total /count
Display Results
end